LESSON GUIDE

Technology Plus

Unit 3: Hardware

1.3.3 – Processing Speed

Lesson Overview:

In this lesson, students will explore processing speed and how it relates to computer performance. They will distinguish between megahertz and gigahertz, practice converting between the two, and compare how these units affect device performance.

Students will:

- Identify and describe processing speed and its role in computer performance.
- Distinguish between megahertz (MHz) and gigahertz (GHz) and convert between the two.
- Analyze device specs and use core counts to decide which device is best for different tasks.

Guiding Question: How does processing speed impact computer performance?

Suggested Grade Levels: 8-12

Technology Needed: No

Approximate Time Required: 45-65 minutes

Standards:

CompTIA Tech+ FC0-U71 Objective

- 1.3 – Compare and contrast common units of measure.
  - Processing speed
    - Megahertz (MHz)
    - Gigahertz (GHz)

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Lesson Background

Background Information:

Processing speeds measure how quickly a computer's CPU can complete cycles of instructions. One hertz (Hz) represents one cycle per second. Processors measure speed in either megahertz (MHz) or gigahertz (GHz). MHz = one million cycles per second, while GHz = one billion cycles per second. Modern CPUs generally operate in the GHz range, but MHz is still used for older hardware or certain computing components. A CPU's clock speed measures how many cycles it can complete per second. Each cycle is a tiny step in processing instructions. An electrical square wave is used as the CPUs clock signal, letting it know when to start the next cycle. The frequency of the square wave determines the CPU's clock speed. Modern CPUs often have multiple cores which allow them to perform several tasks simultaneously. Clock speed determines how fast each core works, while the number of cores determines how many tasks the CPU can handle at once.

Materials Needed:

Materials per Class	• Lesson Guide • Lesson Slides 1.3.3 – Processing Speed • Activity 1.3.3 – Hertz Conversion and Tech Specs
Materials per Student	• Student Notes 1.3.3 – Processing Speed • Student Handout 1.3.3 – Processing Speed • Assessment 1.3.3 – Processing Speed (optional) • Calculator

Cyber Terms:

Term	Definition
processing speed	how fast a computer's central processing unit (CPU) can carry out instructions, usually measured in hertz (Hz)
clock speed	the rate at which a CPU processes instructions, measured in hertz (Hz); each "tick" of the clock signals the CPU to perform the next step of its work
square wave	an electrical signal that switches back and forth between two levels (high and low) in a square-like shape; CPUs use square waves for timing, where each change (tick) represents the next step of work
core (CPU)	a processing unit inside the CPU that can run its own tasks independently; modern CPUs often have multiple cores, which lets them handle more tasks at the same time

1.3.3 – Processing Speed

Guiding Question: How does processing speed impact computer performance?

Lesson Launch (5 minutes)

- **Say:** "Today we will explore how processing speed relates to computer performance, learn about the two main measurement units (megahertz and gigahertz), and compare how clock speed and cores affect a device's performance."
- **Ask:** "What do you think affects how fast your laptop, phone, or gaming console runs?"
- Allow students to share a few quick ideas.
- **Say:** "Let's use a metaphor to understand processing speed. Imagine delivery trucks:
  - If two trucks drive at the same speed, they'll deliver the same number of packages in the same time.
  - But if one company has more trucks, they can deliver more packages in the same amount of time.
  - Clock speed is like how fast the trucks drive. Cores are like how many trucks you have."
- **Ask:** "So, what do you think happens if a CPU has both faster trucks and more trucks?"

  Students may respond: "It gets more done at the same time." or "It's faster overall."
- Display the Tech Specs Sheet and say: "Here's a spec sheet comparing real devices: laptops, a phone, and a Raspberry Pi model. Notice the Clock Speed (GHz or MHz) and the number of Cores. These numbers tell us how fast and how many tasks each device can handle at once."
- **Ask:** "Which device has the fastest clock speed? Which has the most cores?"

  Students may point out: Smartphone C has 3.2 GHz and 8 cores.
- **Say:** "When we put clock speed and cores together, we get a clearer picture of how powerful a CPU really is. Today, we're going to dig into how processing speed works and why it matters for device performance."

Clocking In: Understanding Processing Speed (20-25 minutes)

- Distribute the Student Handout 1.3.3 – Processing Speed and have students fill in the answers during the lesson.
- Introduce ***processing speed***, its definition, hertz, and high processing possibilities.
  - Hertz Examples: If you stomp your foot once every second, you are stomping at one Hz per second. If you stomp ten times every second, you are stomping at 10Hz.
  - Although faster is better, fast CPUs generate more heat. To save power when it's not busy, CPU's will throttle down their speed to reduce the amount of energy used.

- Introduce ***clock speed*** and ***square wave***, their definitions, and how they relate to processing speeds.
  - A higher clock speed allows the CPU to complete more cycles per second, which means faster processing.
  - The square wave goes up and over then down and over in one cycle.
  - Example: Imagine a marching band that only steps forward when the drum hits. The drumbeat is the square wave, or clock signal, and the number of beats per second is the clock speed. The sharper and more consistent the beat, the easier it is for the band (CPU) to stay in sync.
- Introduce megahertz (MHz) and gigahertz (GHz), their capacity, and examples.
- Discuss conversion rules for megahertz and gigahertz.
- Introduce CPU ***cores*** and how they impact processing speeds.
  - Some programs can only use one core, so high GHz matters more.
  - Some programs can use all cores, so having more cores is better even if each one is slightly slower.
  - Example:
    - Clock speed = how fast each chef makes a pizza.
    - Cores = how many chefs you have.
    - Having one fast chef is great for a single pizza order, but for an order of ten pizzas having four chefs working at the same time is much better.
  - Cores are discussed more in lesson 2.2.1 – Internal Computing Components.

Activity: Hertz Conversion and Tech Specs (15-25 minutes)

- Display Activity Slides 1.3.3 – Hertz Conversion and Tech Specs, have students record their answers in the handout.
- Choose if you want students to work together or in a small group.
- Students will convert between the two processing units.
- Students will view a simplified tech specs sheet.
  - Convert each device to the same clock speed.
  - Identify which device has the fastest single core speed.
  - Calculate potential computing power for each device.
  - Explain which device would be best for:
    - Gaming
    - Office work
    - DIY electronics projects

Putting It All Together (5-10 minutes)

- Review what students learned today by discussing the questions below as a class.
- What is processing speed and how does it impact a computer's performance?

  Processing speed is how fast a computer's central processing unit can carry out instructions, it is measured in Hertz. High processing speed in CPUs can handle more instructions in less time, make programs open faster, allow smoother multitasking, and improve how quickly tasks finish.

- How does clock speed and square wave relate to processing speed?

  Clock speed synchronizes the CPUs timing for every operation; each tick signals the CPU to move to the next step of its work. Square wave is the electrical signal inside the CPU that represents binary digits 1 and 0, its sharp edges make it easy for the CPU to detect the tick signals for each step of the CPUs work.

- What is processing speed measured in and what do they represent?

  Processing speed is measured in hertz, specifically megahertz and gigahertz.

  Megahertz (MHz) = 1,000,000 cycles per second

  Gigahertz (GHz) = 1,000,000,000 cycles per second

- How do cores impact CPU performance?

  A single core CPU can only handle one task at a time, while multi-core CPUs can handle multiple tasks at once. Multi-core processing is helpful for gaming, video editing, running many programs at once, and scientific calculation or simulations.